

Proposing Two Research Ideas to Leverage Speedrun Video Repositories for Diverse Stakeholders

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Meeting Purpose

- Define Our Research Direction and Set Specific Experiments

Flow of the Meeting

1. We (Kyushu-side) present our research ideas.
2. Prof. Cor-Paul provides feedback on our ideas.
3. Based on the feedback, we finalize the research direction and experimental plan.

Why Focus on Speedruns?

- Bugs used in speedruns are reproducible.
- This reproducibility provides a reliable source of real-world bug instances.

Who Benefits from Reproducible Speedrun Bugs?

- **General Public:** Provides entertainment and enjoyment.
- **Active Players:** Serves as a reference for intentionally triggering bugs.
- **Game Developers:** Helps prioritize bug fixes based on the frequency of exploitation.

1. Automated Organization of Optimal Speedrun Chart Evolution

Background: Speedrunning is an entertaining activity enjoyed by many viewers. However, beginners often struggle to understand which bugs and techniques are being used. Charts represent these strategies, and analyzing them could help make speedruns more accessible and enjoyable.

Aim: Enable even beginners to deeply enjoy speedruns with the help of explanations.

Ideal: Automatically analyze various speedrun charts and present them in a museum-style format with explanations.

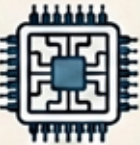
Workflow



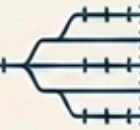
1. Category Selection: Define specific game category (e.g., Super Mario 64, 120 Star).



2. Data Collection: Automatically gather historical world records for a selected category from speedrun.com.



3. Bug Detection (Research Focus): An LLM analyzes each world record to identify the type of exploited bug, its location, and its start and end timestamps.



4. Chart Organization: Automatically organize the charts of each world record based on the identified buggy intervals.



5. Museum Exhibition: The organized charts will become exhibits in a game museum.

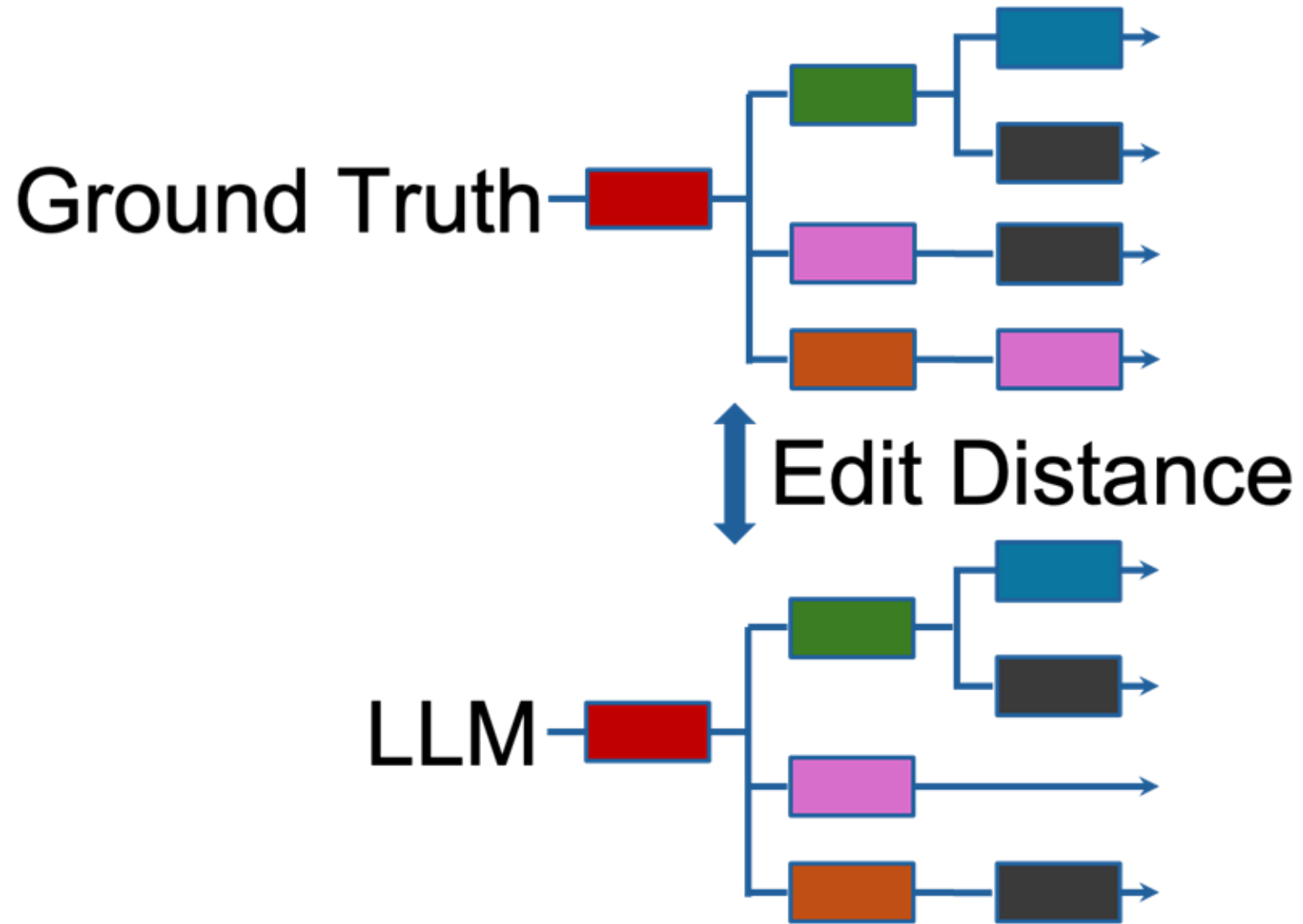
RQs

RQ1: Can an LLM accurately detect buggy intervals in speedrun videos?

RQ2: Can we automatically organize optimal speedrun charts based on detected bug segments?

RQ3: Do the organized charts improve the understanding and enjoyment of speedruns for users?

RQ2: Evaluation Metrics



2. Automated Generation of Bug Highlight Clips

Background: Exploiting bugs is essential for mastering speedruns. Runners study bugs from existing videos, but the number and length of these videos can be overwhelming. We aim to support this process.

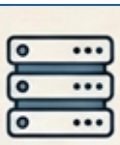
Aim: Support speedrunners in studying and practicing (with potential benefits for developers and viewers).

Ideal: Given a buggy segment with a specific type of bugs, automatically generate clips that contain only the specific type of bugs.

Workflow



1. **Game & Bug Selection:** Define specific game (e.g., Super Mario 64) and bug (e.g. Backwards Long Jump - BLJ).



2. **Data Collection:** Automatically gather the current world records for each category of a selected game from speedrun.com.



3. **Bug Detection (Research Focus):** An LLM analyzes each world record to identify the start and end timestamps of exploited bugs.



4. **Generate Bug Highlight Clips:** Generate bug highlight clips based on the identified buggy intervals.

5. **Benefits for Diverse Stakeholders:**



General Public: Provides entertainment & enjoyment.

Active Player: Facilitates execution study & practice.

Game Developers: Helps prioritize bug fixes by frequency.

RQs

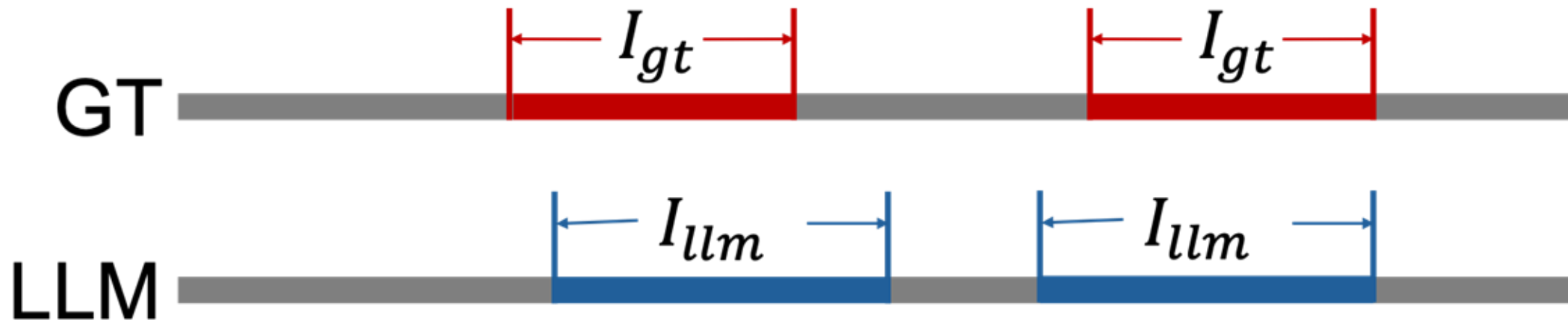
RQ1: How accurately can an LLM identify buggy intervals in speedrun videos?

RQ2: How effectively can bug highlight clips be generated from detected intervals?

RQ3: Do the generated clips support learning and practice for speedrunners?

RQ2: Evaluation Metrics

$$tIoU = \frac{|I_{gt} \cap I_{llm}|}{|I_{gt} \cup I_{llm}|}$$



Anticipated Questions

- How are charts constructed from detection results?
- Connection between the "Why" (Slide 5) and our approaches?